

If AI data centers require fresh water to cool, why don't they make them in cold places or space?

Posted by u/nocumprincess • 0 points • 11 comments

Read somewhere on social media that china made a data center underwater to keep the data center cool and it made a lot of sense. So why are we wasting our precious fresh water resources in keep data centers cool? Why can't they use dirty water? Or make them in cold climates? Or even space?

Comments

u/DanteRuneclaw • 1 points • 2026-03-08 18:11 UTC

Space turns out to be a particularly bad idea. Space isn't really cold in the "there's cold molecules there that you can easily transfer excess heat into" sense. It's more cold in the "there's hardly any molecules there at all" sense.

u/Delicious_Toad • 4 points • 2026-03-08 17:21 UTC

>Why can't they use dirty water?

They're using evaporative cooling. Water removes heat as it evaporates. That means any impurities in the water are concentrated. If you use dirty water, the system will quickly become clogged with sludge, and you'll have to shut it down more frequently for cleaning.

>Or make them in cold climates?

This actually is being done, to some extent. It's not practical to build data centers in sparsely-populated arctic regions because the infrastructure doesn't exist to support them, but there are companies that prefer to site their data centers in, for example, Scandinavian countries, in part because the lower ambient temperature reduces cooling costs.

There are just other factors as well. For example, Texas has been attracting data centers by creating a really friendly regulatory environment. It's not an obviously appealing place to site stuff that needs to be kept cool, since it's a hot and fairly arid state, but if they let you do whatever the hell you want and look the other way when you violate the flimsy rules that do exist, then it can still make financial sense.

>Or even space?

Cooling things in space isn't as easy as you might think. People think of space as 'cold,' and in a sense

that's definitely true. However, it's not like being immersed in a dense, cold fluid: it's a vacuum.

The vacuum of space is technically way colder than, for example, a pond of liquid water. However, if you were to heat up an iron ball until it was red hot, then toss it either into the vacuum of space or into a pond, it would initially cool down WAY faster in the pond—because it would be shedding heat through conduction into the surrounding water, and that heat would be quickly carried away from the ball by the convection that occurs when you immerse a heat source in a fluid.

The ball would stop cooling down in the pond once it equalized with the ambient temperature of the water, so it would ultimately reach a much lower temperature in space, but it would take much longer to get there.

Here's a fun fact: while space suits need heating elements in the extremities—especially the gloves, where the high surface area to volume ratio means astronauts lose a lot of heat through thermal radiation—they actually do more cooling than heating overall. Radiative cooling on its own doesn't remove enough heat from the body and integrated electronics to keep the astronauts inside from overheating, so they use systems that transfer heat into water and then eject that into space.

u/LeeBucketsEatsOrAnGe • 1 points • 2026-03-13 21:23 UTC

I don't understand why shutting them down more often for cleaning is a problem. They have the resources to do that and draining clean water doesn't seem good enough of a reason when they can just shut down the system every week

u/Andeol57 • 1 points • 2026-03-08 16:14 UTC

Not that it's really the issue here, but space isn't all that great to cool things off.

Cooling off requires taking thermal energy from the object and sending it elsewhere. Typically, you transfer that energy from the object to cold water, or air, or whatever. But in space, what are you transferring that heat to? There is nothing around you.

You can compare that to cooling things down with air or water. Water is fast, because it transmits heat very well. Air is much slower. So even lukewarm water will often cool things off faster than cold air. Well emptiness is even much slower than air, no matter how "cold" it is.

Stuff can still get cooler in space just by emitting light (typically infrared), but that's not a particularly fast way to cool down. In a way, that's lucky. It means maintaining a space station at living temperature isn't consuming much energy.

u/DiamondJim222 • 2 points • 2026-03-08 16:14 UTC

Getting things into space is very expensive. You also need massive amounts of power, much more than the solar panels on satellites can generate. Finally - the information transmission rates from space are too slow to be functionally useful.

u/DiogenesKuon • 1 points • 2026-03-08 15:49 UTC

Because the water usage argument is massively overstated, and not really a concern for the data centers themselves, as the cost to acquire the water is negligible to them. It only becomes a problem if local water supply is low and government starts limiting usage.

u/KronusIV • 2 points • 2026-03-08 15:12 UTC

Space, the space we can actually reach fairly easily, is pretty hot. If you're in space and exposed to sunlight you're taking on a lot of heat. The ISS has to really work to dump its heat. And even if you kept

your center in the shade somehow dumping heat via radiative cooling is the slowest method. You'd need thermal fins the size of football fields to have even a chance.

u/Long-Aardvark-3129 • 1 points • 2026-03-08 14:40 UTC

Money.

u/Nine_Eighty_One • 3 points • 2026-03-08 14:15 UTC

I'm not sure how the proposed space-based data centers are supposed to work but cooling is known to be a problem in space. The common misconception is that space is cold but it's not like cold air or water, in the void heat loss can only happen through radiation so it happens really slowly.

u/FriendOfDorian • 1 points • 2026-03-08 14:10 UTC

This is just a guess because i don't know anything about the data centres and how they run but my guess is, it's quite difficult to get to space or under the ocean. So building the centres, making adjustment and also maintenance would be incredibly difficult. It may be more efficient in the long term but business's investing in AI don't really care about long term. It's about appeasing investors and making money for the right now.

u/generic_redditor_71 • 10 points • 2026-03-08 14:08 UTC

Data centers need cooling but they also need power, data connections, workers, access to services, and they need to be as close as possible to their customers. If you put a data center in the Arctic you're making cooling easier but everything else much more difficult and expensive.

u/Useful_Location_6728 • 2 points • 2026-03-08 14:08 UTC

There's no water in space...